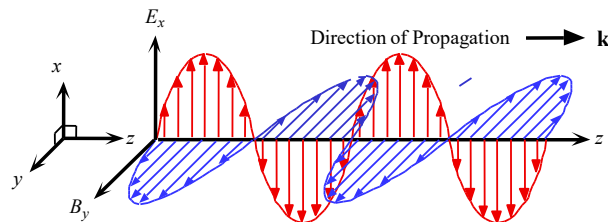


Types of Transmission Modes

- Normally the term “Transmission line” is used for the EM waveguiding structure that is capable of guiding TEM (Transverse Electric and Magnetic) waves.
- TEM waves can only exist in structures that contain two or more separate conductors to carry EM signals.**



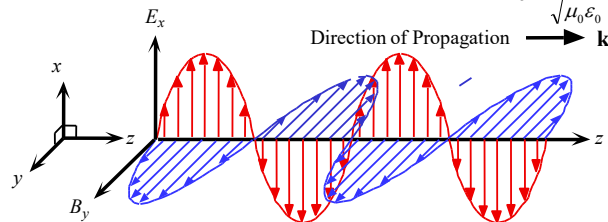
Any TEM wave is a travelling wave which has time varying electric and magnetic fields which are perpendicular to each other and the direction of propagation.

1

Types of Transmission Modes

- The magnetic fields that change with time induce electric fields as explained by Faraday's law, and the time-varying electric fields induce magnetic fields, as explained by the generalized Ampere's law (We will deal with these two laws later on to develop the Maxwell's equations)
- This interchange of energy causes that EM wave travels with speed of light in free space which fundamentally is related to free space electric permittivity and magnetic permeability, namely

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

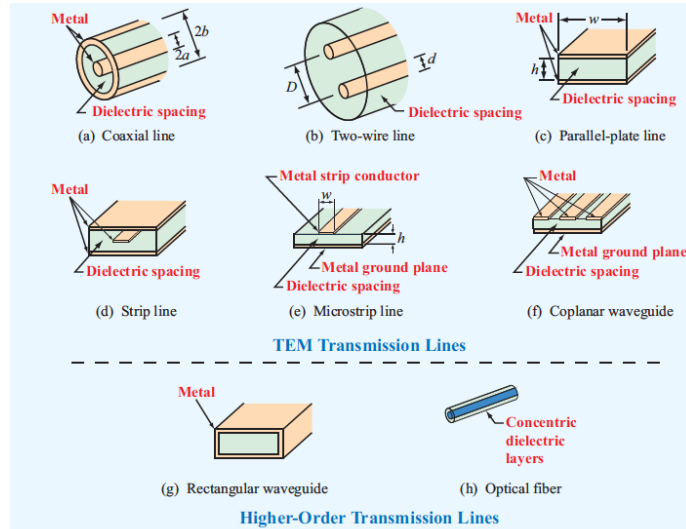


Any TEM wave is a travelling wave which has time varying electric and magnetic fields which are perpendicular to each other and the direction of propagation.

2

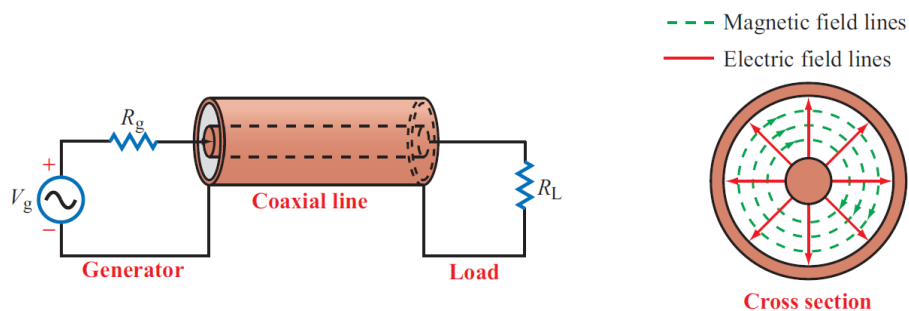
Types of Transmission Modes

TEM (Transverse Electromagnetic):
Electric and magnetic fields are orthogonal to one another, and both are orthogonal to direction of propagation



3

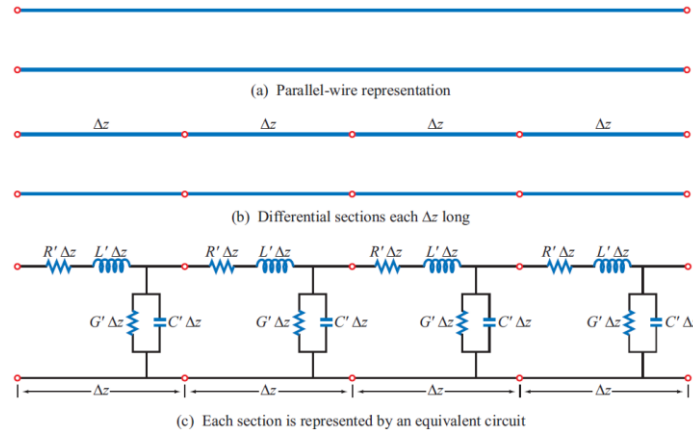
Example of TEM Mode



Electric Field E is radial
Magnetic Field H is azimuthal
Propagation is into the page

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Transmission Line Model



- R' : The combined **resistance** of both conductors per unit length, in Ω/m .
- L' : The combined **inductance** of both conductors per unit length, in H/m .
- G' : The **conductance** of the insulation medium between the two conductors per unit length, in S/m .
- C' : The **capacitance** of the two conductors per unit length, in F/m .

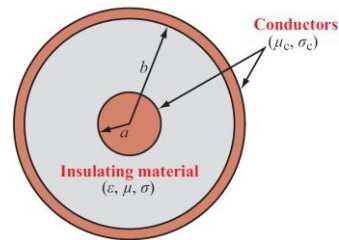
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Table 2-1: Transmission-line parameters R' , L' , G' , and C' for three types of lines.

Parameter	Coaxial	Two-Wire	Parallel-Plate	Unit
R'	$\frac{R_s}{2\pi} \left(\frac{1}{a} + \frac{1}{b} \right)$	$\frac{2R_s}{\pi d}$	$\frac{2R_s}{w}$	Ω/m
L'	$\frac{\mu}{2\pi} \ln(b/a)$	$\frac{\mu}{\pi} \ln \left[(D/d) + \sqrt{(D/d)^2 - 1} \right]$	$\frac{\mu h}{w}$	H/m
G'	$\frac{2\pi\sigma}{\ln(b/a)}$	$\frac{\pi\sigma}{\ln \left[(D/d) + \sqrt{(D/d)^2 - 1} \right]}$	$\frac{\sigma w}{h}$	S/m
C'	$\frac{2\pi\epsilon}{\ln(b/a)}$	$\frac{\pi\epsilon}{\ln \left[(D/d) + \sqrt{(D/d)^2 - 1} \right]}$	$\frac{\epsilon w}{h}$	F/m

Notes: (1) Refer to Fig. 2-4 for definitions of dimensions. (2) μ , ϵ , and σ pertain to the insulating material between the conductors. (3) $R_s = \sqrt{\pi f \mu_c / \sigma_c}$. (4) μ_c and σ_c pertain to the conductors. (5) If $(D/d)^2 \gg 1$, then $\ln \left[(D/d) + \sqrt{(D/d)^2 - 1} \right] \approx \ln(2D/d)$.

The pertinent **constitutive parameters** apply to all three lines and consist of two groups: (1) μ_c and σ_c are the magnetic permeability and electrical conductivity of the conductors, and (2) ϵ , μ , and σ are the electrical permittivity, magnetic permeability, and electrical conductivity of the insulation material separating them.



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Example 2-1: Air Line

An *air line* is a transmission line in which air separates the two conductors, which renders $G' = 0$ because $\sigma = 0$. In addition, assume that the conductors are made of a material with high conductivity so that $R' \simeq 0$. For an air line with a characteristic impedance of 50Ω and a phase constant of 20 rad/m at 700 MHz , find the line inductance L' and the line capacitance C' .

Solution: The following quantities are given:

$$\begin{aligned} Z_0 &= 50 \Omega, & \beta &= 20 \text{ rad/m}, \\ f &= 700 \text{ MHz} = 7 \times 10^8 \text{ Hz}. \end{aligned}$$

With $R' = G' = 0$, Eqs. (2.25b) and (2.29) reduce to

$$\begin{aligned} \beta &= \Im \left[\sqrt{(j\omega L')(j\omega C')} \right] \\ &= \Im \left(j\omega \sqrt{L'C'} \right) = \omega \sqrt{L'C'}, \\ Z_0 &= \sqrt{\frac{j\omega L'}{j\omega C'}} = \sqrt{\frac{L'}{C'}}. \end{aligned}$$

The ratio of β to Z_0 is

$$\frac{\beta}{Z_0} = \omega C',$$

or

$$\begin{aligned} C' &= \frac{\beta}{\omega Z_0} \\ &= \frac{20}{2\pi \times 7 \times 10^8 \times 50} \\ &= 9.09 \times 10^{-11} \text{ (F/m)} = 90.9 \text{ (pF/m)}. \end{aligned}$$

From $Z_0 = \sqrt{L'/C'}$, it follows that

$$\begin{aligned} L' &= Z_0^2 C' \\ &= (50)^2 \times 90.9 \times 10^{-12} \\ &= 2.27 \times 10^{-7} \text{ (H/m)} = 227 \text{ (nH/m)}. \end{aligned}$$

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Coaxial TL

- The dominant mode is TEM
- For lossless TL ($R = 0$ & $G = 0$)

$$-Z_o = \sqrt{\frac{L'}{C'}} = \frac{\ln(b/a)}{2\pi} \sqrt{\frac{\mu_o}{\epsilon_o \epsilon_r}}$$

$$Z_o = \frac{60}{\sqrt{\epsilon_r}} \ln\left(\frac{b}{a}\right)$$

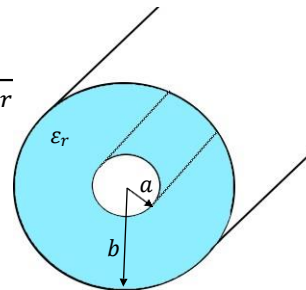
$$-\alpha = 0 \text{ \& } \beta = \omega \sqrt{L'C'} = \omega \sqrt{\mu_o \epsilon_o \epsilon_r}$$

$$\beta = \frac{\omega}{c_o} \sqrt{\epsilon_r} \quad c_o = \frac{1}{\sqrt{\mu_o \epsilon_o}} = 3 \times 10^8 \text{ m/s}$$

$$-v_{ph} = \frac{\omega}{\beta}$$

$$v_{ph} = \frac{c_o}{\sqrt{\epsilon_r}}$$

Parameter	Coaxial
R'	$\frac{R_s}{2\pi} \left(\frac{1}{a} + \frac{1}{b} \right)$
L'	$\frac{\mu}{2\pi} \ln(b/a)$
G'	$\frac{2\pi\sigma}{\ln(b/a)}$
C'	$\frac{2\pi\epsilon}{\ln(b/a)}$



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Coaxial TL

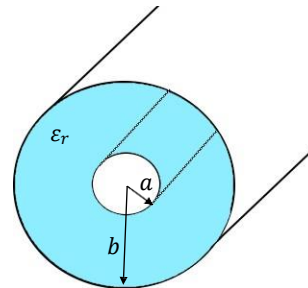
- Guided Wavelength

$$-\lambda_g = \frac{2\pi}{\beta} \quad \& \quad \lambda_o = \frac{c_o}{f}$$

$$\lambda_g = \frac{\lambda_o}{\sqrt{\epsilon_r}}$$

The first higher-order mode is TE₁₁, with cutoff frequency f_c

$$f_c = \frac{1}{\pi(a+b)\sqrt{\mu_r\epsilon_r}}$$



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Propagation through coaxial mediums



$$Z_o = \frac{138.2}{\sqrt{\epsilon}} \log\left(\frac{b}{a}\right)$$

Example 3.5 mm connector

Outer diameter of conductor:

$$b = 3.50 \text{ mm}$$

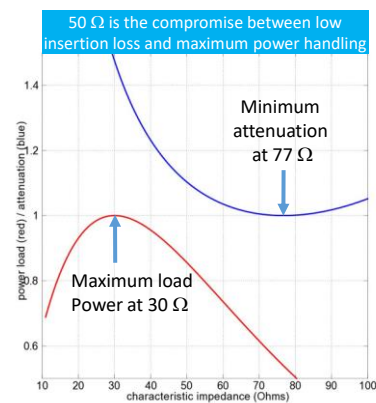
Inner diameter of conductor:

$$a = 1.52 \text{ mm}$$

Dielectric constant of insulator:

$$\epsilon = 1$$

$$Z_o = \frac{138.2}{\sqrt{1}} \log\left(\frac{3.50}{1.52}\right) = 50 \Omega$$



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Transmission line characteristic Impedance

Z_L	b/a	Special Characteristics	Application
77 Ω	3.6	min. attenuation	TV-receiving-antenna-line
60 Ω	2.72	max. voltage capability	No special use
30 Ω	1.65	Best power loading capacity	For very high transmitter-power
150 Ω	13	min. capacity per unit length	Low capacity test-lines (sensing head)
50 Ω	2.3	Compromise between α_{min} and P_{max}	Universally normed, RF measurement technique
93 Ω	4.8	Compromise between α_{min} and C_{min}	Lines used in data-communications

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Coaxial Cables in reality



iEMSL Video Tutorial: https://www.youtube.com/watch?v=-o6a_VJpf4

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Example 1

Design a coaxial transmission line and find the characteristic impedance with a cutoff frequency of $f_c = 5GHz$, built with a dielectric material with relative permittivity of $\epsilon_r = 2.5$, and an inner thickness of $a = 2.5mm$

$$f_c = \frac{1}{\pi(a+b)\sqrt{\mu_o\epsilon_o\epsilon_r}}$$

$$Z_0 = \frac{\ln(b/a)}{2\pi} \sqrt{\frac{\mu_o}{\epsilon_o\epsilon_r}}$$

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Example 1

Design a coaxial transmission line and find the characteristic impedance with a cutoff frequency of $f_c = 5GHz$, built with a dielectric material with relative permittivity of $\epsilon_r = 2.5$, and an inner thickness of $a = 2.5mm$

- $f_c = \frac{1}{\pi(a+b)\sqrt{\mu_o\epsilon_o\epsilon_r}}$
- $f_c(a+b) = \frac{1}{\pi\sqrt{\mu_o\epsilon_o\epsilon_r}}$
- $b = \frac{1}{\pi f_c \sqrt{\mu_o\epsilon_o\epsilon_r}} - a$
- $b = 0.00957m = 9.57mm$
- $Z_0 = \frac{\ln(7/6)}{2\pi} \sqrt{\frac{\mu_o}{\epsilon_o\epsilon_r}}$
- $Z_0 = 50.92\Omega$

$$f_c = \frac{1}{\pi(a+b)\sqrt{\mu_o\epsilon_o\epsilon_r}}$$

$$Z_0 = \frac{\ln(b/a)}{2\pi} \sqrt{\frac{\mu_o}{\epsilon_o\epsilon_r}}$$

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Example 2 : RG-59

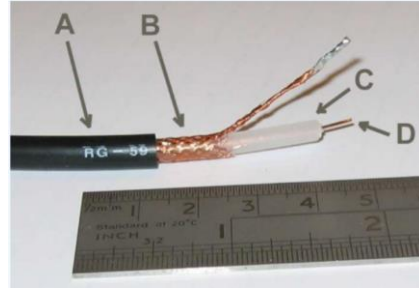
- $a \approx 0.292 \text{ mm}$ $b \approx 1.855 \text{ mm}$
- The dielectric material is polyethylene having
 - $\epsilon_r \approx 2.25$
 - $\sigma_s \approx 5.9 \times 10^{-5} \text{ S/m}$
 - resistance per unit length $R = 0.1 \Omega / \text{m}$

Find

- L' , C' and G' ?
- Does this cable satisfy low-loss criteria for $f = 1 \text{ GHz}$?

$$R' \ll \omega L' \quad G' \ll \omega C'$$

- Z_o , β , v_{ph} , λ_g
- α (hint $\alpha = \text{real}\{\gamma\}$)



RG-59 coaxial line. A: Insulating jacket. B: Braided outer conductor. C: Dielectric spacer. D: Inner conductor

Parameter	Coaxial
R'	$\frac{R_s}{2\pi} \left(\frac{1}{a} + \frac{1}{b} \right)$
L'	$\frac{\mu}{2\pi} \ln(b/a)$
G'	$\frac{2\pi\sigma}{\ln(b/a)}$
C'	$\frac{2\pi\epsilon}{\ln(b/a)}$

$$Z_o = \frac{60}{\sqrt{\epsilon_r}} \ln\left(\frac{b}{a}\right)$$

$$\beta = \frac{\omega}{c_o} \sqrt{\epsilon_r}$$

$$v_{ph} = \frac{c_o}{\sqrt{\epsilon_r}}$$

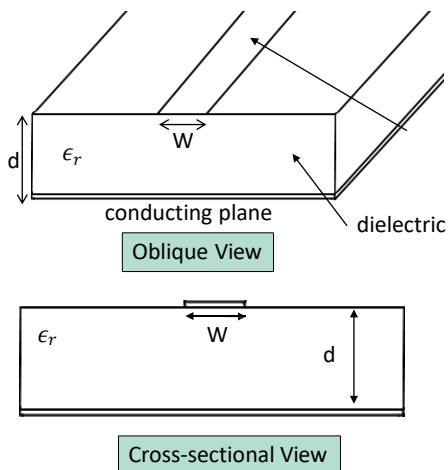
$$\lambda_g = \frac{\lambda_o}{\sqrt{\epsilon_r}}$$

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Microstrip line

- Used in all PCB lines
- Very simple to manufacture
- Dominant mode Quasi-TEM
- How to calculate L , C , R , G and hence characteristic impedance and propagation constant???
 - Empirical formulas derived from data obtained from measurements
 - Quasi Static method: Solve at DC and Claim it is valid at high frequency
 - Rigorous numerical technics based on Maxwell's equation



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Microstrip line

- For lossless TL :

The characteristic impedance Z_0 is given by:

$$Z_0 = \begin{cases} \frac{60}{\epsilon_e} \ln \left(\frac{8d}{w} + \frac{w}{4d} \right) & , \quad \frac{w}{d} \leq 1 \\ \frac{120\pi}{\sqrt{\epsilon_e} \left[\frac{w}{d} + 1.393 + 0.667 \ln \left(\frac{w}{d} + 1.444 \right) \right]} & , \quad \frac{w}{d} \geq 1 \end{cases}$$

and the phase velocity is:

$$v_{ph} = \frac{1}{\sqrt{\mu_0 \epsilon_e \epsilon_r}}$$

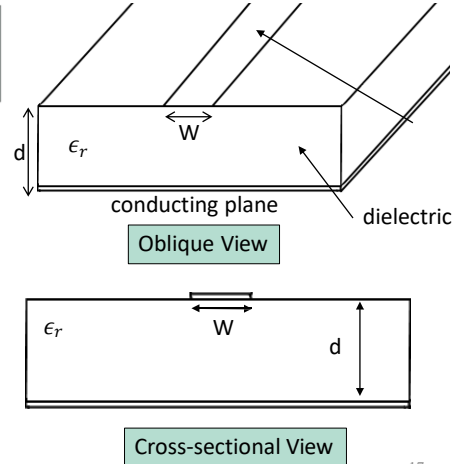
where ϵ_e is given by:

$$\epsilon_e = \frac{\epsilon_r + 1}{2} + \frac{\epsilon_r - 1}{2} \frac{1}{\sqrt{1 + 12 \frac{d}{W}}}$$

$$\beta = \frac{\omega}{c_0} \sqrt{\epsilon_e}$$

$$v_{ph} = \frac{c_0}{\sqrt{\epsilon_e}}$$

$$\lambda_g = \frac{\lambda_0}{\sqrt{\epsilon_e}}$$



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Microstrip - Frequency Dependency

- So far, we defined the effective permittivity at DC we called it ϵ_e or $\epsilon_e(0)$ and the characteristic impedance at DC is Z_0 or $Z_0(0)$. These are also called the quasi-static effective permittivity and quasi-static characteristic impedance.

$$\epsilon_e(f) = \epsilon_r - \frac{\epsilon_r - \epsilon_e(0)}{1 + (f/f_a)^m}$$

$$Z_0(f) = \frac{Z_0(0) \sqrt{\epsilon_e(0)}}{\sqrt{\epsilon_e(f)}}$$

$$\begin{aligned} f_a &= \frac{f_b}{0.75 + (0.75 - 0.332\epsilon_r^{-1.73})(w/h)} \\ f_b &= \frac{47.746 \times 10^6}{h \sqrt{\epsilon_r - \epsilon_e(0)}} \tan^{-1} \left\{ \epsilon_r \sqrt{\frac{\epsilon_e(0) - 1}{\epsilon_r - \epsilon_e(0)}} \right\} \\ m &= \begin{cases} m_0 m_c & m_0 m_c \leq 2.32 \\ 2.32 & m_0 m_c > 2.32 \end{cases} \\ m_0 &= 1 + \frac{1}{1 + \sqrt{w/h}} + 0.32 \left(1 + \sqrt{w/h} \right)^{-3} \\ m_c &= \begin{cases} 1 + \frac{1.4}{1 + w/h} \{ 0.15 - 0.235 e^{-0.45 f/f_a} \} & \text{for } w/h \leq 0.7 \\ 1 & \text{for } w/h > 0.7 \end{cases} \end{aligned}$$

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Class activity

- For $\epsilon_r=2.1$ and d (height)= 5 mm design a microstrip line such that $Z_o = 50 \text{ ohm}$

Calculation

Dielectric Constant: 2.1

Width: 1 ??? Millimeters

Height: 5 Millimeters

CALCULATE

Result:

Width/Height: 0.2000

Effective Dielectric Constant: 1.635

Impedance: 173.2 Ω ← $Z_o = 50 \text{ ohm}$

Go to: <https://www.pasternack.com/t-calculator-microstrip.aspx>

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